



The Future Of The Industry



"The IT Security sector is growing at a phenomenal rate. It is the fastest-growing IT sector in the world in many of the advanced nations. India is soon expected to among the top ten security spenders"

—Kartik Shahani

Regional Director - India
McAfee, Inc



"From a business standpoint, the compelling pressures of complying with law and standards and showing business value and cost-effectiveness for security measures will be one of the main challenges in the coming years"

—K S Ganesan

CTO, Microland Ltd



"Solutions with proactive blocking capabilities are gaining traction in India and are expected to overtake traditional Intrusion Detection solutions in the near future"

—Lokesh Mehra

CTO and Regional Manager -
Corporate Responsibility
Cisco - South Asia



"Five to 10 years down the line, when all data will be networked, it will be a big challenge to keep it secure. This will lead to a huge demand for security professionals"

—Manish Kumar

CEO
Rooman Technologies



"Security will become all-pervasive, and the goal is for it to be non-intrusive. From a technology standpoint, it will be deeply embedded in almost all platforms and enterprise applications"

—Niranjan Meka

Head - RSA Development Centre
EMC India Centre of Excellence

To get into the security market, one must have an educational background that covers the basics of networking, C/C++, and scripting languages

System Administrators are involved in security administration and monitoring of networks, along with the application security environment in an organisation. Their role includes implementation of firewalls and monitoring a Virtual Private Network environment on the network perimeter of the organisation.

Desktop Security Administrators manage and deploy anti-virus and anti-spam software, and are involved in patch management and malware support in any organisation where multiple workstations have to be updated with the latest security patches and kept virus-free.

After you have served on the product-led side for six to seven years, you can get into the more vendor-neutral role of the consulting-led approach. This role requires a deep understanding of networking and security concepts implemented across organisations. You'll be in charge of recommending security enhancements to organisations.

Then, after four to five years of consulting, one can move on to specialise in niche segments of security. Functions include Hacker Attack Analysis, Vulnerability Analysis, Ethical Hacking, and Forensic and Penetration Testing.

Enlightenment

To get into the security market, one must have an educational background that covers the basics of networking, C/C++, and scripting languages. These are covered in courses like B.E., B.Tech., BCA and some B.Com. courses which have Computer Science as a graduating subject in a few state universities. Colleges affiliated with the All India Council of Technical Education (A.I.C.T.E.) are a safe bet to start from.

Alternatively, diploma courses on the basics of networking, C/C++, and scripting languages from NIIT, Aptech Computer Education, and Rooman Technologies among others, are also accepted by organisations and will secure a good foundation your future.

These university and diploma courses will help you get your first job as a software engineer / developer in the software industry. Those willing to further their education can go on to complete their Masters degree—as in M.E., M.Tech., and MCA. Since some subjects in M.E., M.Tech., and MCA degree courses teach the principles of networking and security in detail, companies naturally prefer candidates with a Masters degree over those with a Bachelor's degree while recruiting. Even further on in your career, the Master's degree will definitely be an add-on.

Apart from the above, there are industry certifications that you can obtain to add value to your résumé. Some certifications are very vendor-specific, meaning they deal with the security of products created or endorsed by the vendors offering the courses: for example, the Microsoft Certified Professional (MCP) exams that are security-related. Microsoft Certified Systems Engineer (MCSE) with a specialisation in security, Checkpoint Certification as Checkpoint Certified Security Engineer (CCSE), and Cisco Certified Security Professional certifications are more examples. If one were expected to deploy, configure, manage, or architect environments largely encompassing products from a single vendor, these certifications would definitely help: later on in their careers, security professionals in some positions may be expected to know and support products from multiple vendors. For example, even if you have an MCSE certification, the company you work for might ask you to configure a hardware firewall that may be provided by a different vendor.

After serving on the product-led side for six to seven years, you gain enough experience to offer your expertise as a Security Consultant. This role involves you to have a vendor-neutral objective of security. This is because different companies may have differ-